Chapter 1

AN INTRODUCTION TO MICROSCOPY

Microscopy involves the study of objects that are too small to be examined by the unaided eye. In the SI (metric) system of units, the sizes of these objects are expressed in terms of sub-multiples of the meter, such as the **micrometer** (1 μm = 10^{-6} m, also called a *micron*) and also the **nanometer** (1 nm = 10^{-9} m). Older books use the Angstrom unit (1 Å = 10^{-10} m), not an official SI unit but convenient for specifying the distance between atoms in a solid, which is generally in the range 2 – 3 Å.

To describe the wavelength of fast-moving electrons or their behavior inside an atom, we need even smaller units. Later in this book, we will make use of the **picometer** (1 pm = 10^{-12} m).

The diameters of several small objects of scientific or general interest are listed in Table 1-1, together with their approximate dimensions.

Table 1-1. Approximate sizes of some common objects and the smallest magnification M* required to distinguish them, according to Eq. (1.5).

Object	Typical diameter *D*	M* = 75μm / *D*
Grain of sand	1 mm = 1000 μm	None
Human hair	150 μm	None
Red blood cell	10 μm	7.5
Bacterium	1 μm	75
Virus	20 nm	4000
DNA molecule	2 nm	40,000
Uranium atom	0.2 nm = 200 pm	400,000

1.1 Limitations of the Human Eye

Our concepts of the physical world are largely determined by what we *see* around us. For most of recorded history, this has meant observation using the human eye, which is sensitive to radiation within the **visible region** of the electromagnetic spectrum, meaning wavelengths in the range 300 – 700 nm. The eyeball contains a fluid whose refractive index ($n \approx 1.34$) is substantially different from that of air ($n \approx 1$). As a result, most of the refraction and focusing of the incoming light occurs at the eye's curved front surface, the **cornea**; see Fig. 1-1.

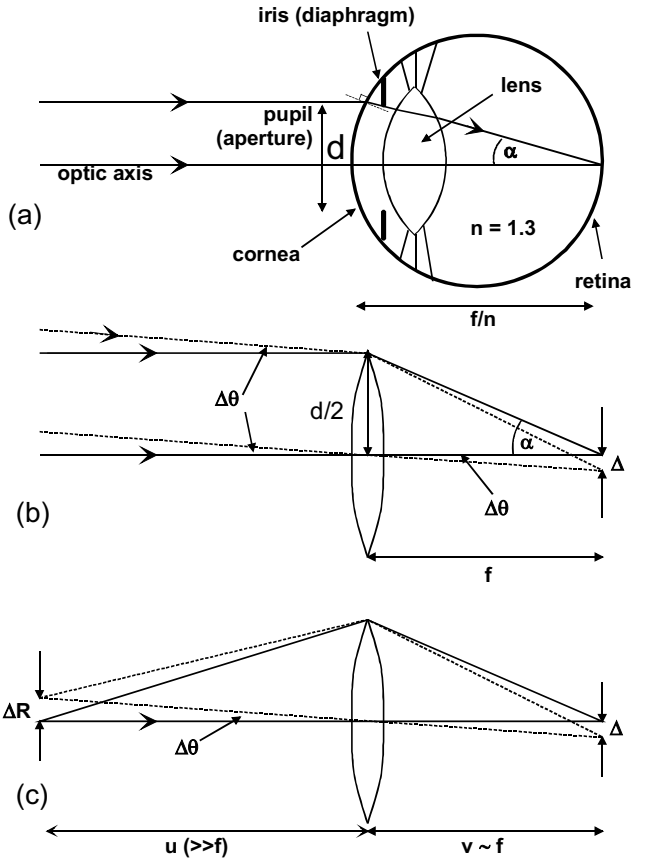


Figure 1-1. (a) A physicist's conception of the human eye, showing two light rays focused to a single point on the retina. (b) Equivalent thin-lens ray diagram for a *distant* object, showing *parallel* light rays arriving from opposite ends (solid and dashed lines) of the object and forming an image (in air) at a distance f (the focal length) from the thin lens. (c) Ray diagram for a *nearby* object (object distance $u = 25$ cm, image distance v slightly less than f).